HOSPITAL MANAGEMENT SYSTEM

Software Defect Report

Console-Based C++ Application — JSON Persistence Layer

Prepared by: Senior QA Analyst & Defect Documentation Specialist

Document Type: Formal Defect Report (FDR)
Version: 1.0
Date: May 2026
Project: Hospital Management System — University Software Testing Course

Contents

1 Executive Summary **2**

2 Defect Summary Table **3**

3 Detailed Defect Reports **4**
3.1 Defects Derived from EXPECT_FALSE Assertions 4
3.2 Functional and Implementation Defects 4

4 Testing Weakness Analysis **5**
4.1 Most Vulnerable Modules 5
4.2 Reliability Concerns 5

5 Risk Assessment **5**
5.1 Overall Risk Rating: MEDIUM-HIGH 5

6 Final QA Evaluation **5**

1 Executive Summary

This Formal Defect Report (FDR) documents all identified defects discovered during the analysis and testing of the Hospital Management System (HMS) — a console-based C++ application with JSON file persistence. The analysis examined the complete source code (**HospitalService**, **HospitalRepository**, **HospitalConsole**), all GTest test suites, the Master Test Case Specification, and the Software Requirements Specification.

The defect identification process covered: (1) conversion of all **EXPECT_FALSE** assertions into formal defect entries, (2) analysis of failing validation scenarios, (3) design and implementation weaknesses, and (4) testing coverage gaps.

Defect Report Summary	
Total Defects Identified	20
Critical Severity	1
High Severity	7
Medium Severity	9
Low Severity	3
Defects with Status: Open	20
Primary Source of Defects	Validation Logic & Console Layer
Modules Affected	5

2 Defect Summary Table

The table below provides a consolidated overview of all documented defects. Detailed entries follow in Section 3.

ID	Title	Module	Sev	Pri	Status
DEF-001	Whitespace-Only Patient Name Accepted	HospServ	High	High	Open
DEF-003	Past Appointment Dates Accepted	HospServ	High	High	Open
DEF-009	Age Boundary Inconsistency (SRS vs Impl)	HospServ	High	High	Open
DEF-010	No Duplicate Patient ID Prevention	HospServ	Crit	Crit	Open
DEF-012	JSON Persistence Silently Fails	HospRepo	High	High	Open
DEF-014	No Input Validation for Alphabetic Age	HospCons	High	High	Open
DEF-017	No GTest Coverage for HospitalConsole	HospCons	High	High	Open

3 Detailed Defect Reports

3.1 Defects Derived from EXPECT_FALSE Assertions

DEF-001 | Whitespace-Only Patient Name Accepted

Description: validatePatientInput() checks only if the name string is non-empty. A name of " " passes. SRS requires rejection of blank-only names.
Steps to Reproduce: 1. Register Patient -> 2. Name: " " -> 3. Age: 30 -> 4. Confirm.
Expected Result: System rejects name and displays validation error.
Actual Result: Registration succeeds. Record stored with blank name.
Root Cause: Missing trim check in validation logic.
Suggested Fix: Add `trimmed.empty()` check after removing whitespace.

DEF-003 | Past Appointment Dates Accepted

Description: validateAppointmentInput() checks format but not if date is in the past. "2020-01-01" is accepted.
Expected Result: Reject past date: "Date must be in the future."
Actual Result: Appointment created for year 2020.
Suggested Fix: Compare epoch value against `std::time()`.

3.2 Functional and Implementation Defects

DEF-010 | No Duplicate Patient ID Prevention

Description: IDs are generated using `nextPatientId++`. If external edits make IDs non-contiguous, collisions occur upon reload.
Expected Result: System prevents creation of duplicate IDs at all times.
Actual Result: ID collision possible on new registrations after loading non-contiguous data.
Root Cause: Initializing `nextId` from `max+1` instead of scanning all IDs.

4 Testing Weakness Analysis

4.1 Most Vulnerable Modules

Module	Defect Count
HospitalService	10
HospitalConsole	6
HospitalRepository	3
Test Infrastructure	1

4.2 Reliability Concerns

- File write failures are silently ignored.
- Corrupted JSON initializes empty database without warning.
- Non-numeric console input causes `std::cin` fail-state.

5 Risk Assessment

5.1 Overall Risk Rating: MEDIUM-HIGH

The system risk is rated High due to critical data integrity risks (ID collisions) and the complete lack of testing for the HospitalConsole layer, which handles all user interactions.

6 Final QA Evaluation

The Hospital Management System shows a professional 3-layer architecture but suffers from incomplete semantic validation and silent persistence failures. Before production, DEF-003 (Past Dates) and DEF-009 (Age Boundary) must be resolved to align with the SRS.

— End of Defect Report —